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Molecular Innovations Shaping Beak Morphology in Birds

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beak morphology, evolution, development, genetics, epigenetics

Abstract

The beak, a pivotal evolutionary trait characterized by high morphological diversity and plasticity, has enabled birds to survive mass extinction events and subsequently radiate into diverse ecological niches worldwide. This remarkable ecological adaptability underscores the importance of uncovering the molecular mechanisms shaping avian beak morphology, particularly benefiting from the rapidly advancing archives of genomics and epigenomics. We review the latest advancements in understanding how genetic and epigenetic innovations control or regulate beak development and drive beak morphological adaptation and diversification over the past two decades. We conclude with several recommendations for future endeavors, expanding to more bird lineages, with a focus on beak shape and the lower beak, and conducting functional experiments. By directing research efforts toward these aspects and integrating advanced omics techniques, the complex molecular mechanisms involved in avian beak evolution and morphogenesis will be deeply interpreted.

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1. INTRODUCTION

The beak stands out as one of the most prominent characteristics evolved by birds. Since Darwin's observations of finches in the Galápagos Islands, beaks have served as a classic trait to illustrate ecological adaptation and evolutionary processes of birds (1–7). Over the span of 165 years, extensive research addressing the question of how diverse bird beak forms evolved has demonstrated the influential role of studies on the avian beak in advancing the theory of natural selection and, more broadly, enhancing our understanding of ecology and evolutionary biology. The rise of evolutionary developmental biology has also granted scientists insights into how this special structure develops in the embryo. However, identifying the specific genes or small molecules responsible for bird beak evolution and development has long been challenging, primarily due to the limitations of applying molecular biology research techniques and methods to birds. Fortunately, over the past two decades, breakthroughs in technologies such as quantitative genetics, transcriptomics, genomics, and epigenomics have significantly enriched our understanding of the molecular mechanisms regulating beak evolution and development. We briefly introduce the nature and development of avian beaks. Then, we focus on recent advances in research on the genetic and epigenetic innovations shaping avian beak morphology. Additionally, we provide suggestions for future research aimed toward extending our understanding of molecular regulations underlying beak morphology.

1.1. Beak Form Diversification Drives the Global Radiation of Birds

Over the past 100 million years (8), the avian tree of life has witnessed an explosion of phenotypic diversity in tandem with the global expansion of birds in both geographical and ecological space (9, 10). This expansion is tied closely to a simultaneous and striking variation in beak morphology (11) (**Figure 1**), a pivotal feature with multifaceted functions in birds. Avian beaks serve various purposes, including but not limited to foraging, preening, nest building, self-defense, communication, and cavity excavation (12–14). Furthermore, the diverse forms of beaks are intricately linked to the array of habitats where birds reside. The ecological forces shaping beak evolution are evident in highly specialized adaptations. For instance, flamingos possess a grooved and horny-toothed beak tailored for filtering plankton (15). Crossbills exhibit a crossed maxilla and mandible, facilitating seed extraction from conifer cones (16). Hummingbirds feature slender and sometimes decurved beaks, enabling them to drink nectar from flowers (17, 18). The intimate connection between ecology and avian beak morphology is underscored by the hypothesis that the beak played a crucial role in birds surviving the Cretaceous–Paleogene (K–Pg) mass extinction event approximately 66 Mya (19), subsequently leading to an ecological radiation across nearly all parts of the Earth (20, 21) (**Figure 1**).

The diversification of beak forms occurred early in the evolutionary trajectory of modern birds, succeeded by a prolonged period dominated by the packing of morphological space (22). In the initial stages of avian diversification, major forms of beaks were established quickly, and differences among lineages remained relatively stable over time (22). Within specific lineages (e.g., passerines), subsequent morphological variation often displays high plasticity (**Figure 1**), facilitating adaptation to novel niches (23, 24) and responding to rapid environmental changes (4), including those induced by human activities (25, 26). For example, approximately 5.7 Mya, the ancestor of Hawaiian honeycreepers colonized the Hawaiian Islands and underwent a rapid radiation into various novel niches (27). Consequently, honeycreeper beak morphology showcases extreme phenotypic diversity, varying from short to long, thin to thick, straight to extremely decurved, and pointed to hooked (27, 28). Another notable example is the adaptive radiation of Darwin's finches in the Galápagos Islands, a more recent phenomenon occurring approximately 1.5 Mya (29). The

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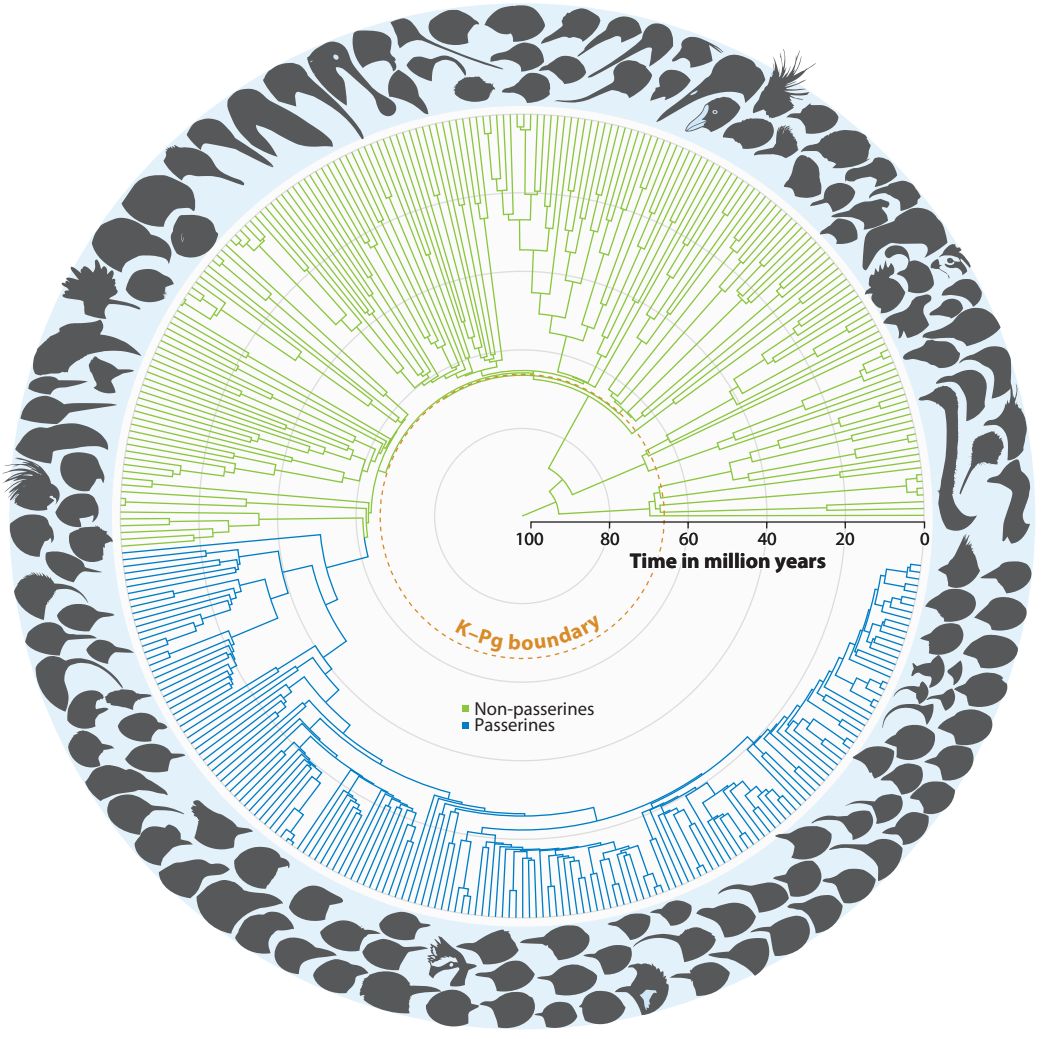

Figure 1

Diverse beak morphology across birds. Birds across the tree of life have undergone an explosion of beak morphological diversity spanning or following the Cretaceous–Paleogene (K–Pg) mass extinction event approximately 66 million years ago (Mya). Almost all major lineages exhibit unique beak forms, with considerable plasticity in beak morphology within lineages, especially within passerines. The family-level time-calibrated phylogeny (**https://erda.ku.dk/archives/341f72708302f1d0c461ad616e783b86/B10K/data_upload/04_timetree/calibration_subset.tre**) involved 363 bird species adapted from a recently published study (21). Silhouette images were adapted from an online open database (PhyloPic, **http://phylopic.org**) without restrictions on reuse (CC0 1.0). All silhouettes were illustrated without scales.

morphological distinctions among these taxa result primarily from variations in beak and body sizes, enabling them to occupy different feeding niches within a confined geographic space where variable seed availability is influenced by severe interannual rainfall fluctuations (4, 30). Additionally, anthropogenic influences have driven adaptive change in beak length, as demonstrated by the well-studied great tit (*Parus major*) (26). Ongoing selection due to the widespread use of artificial feeders since the nineteenth century has led to the evolution of longer beaks in great tits in the United Kingdom (31).

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1.2. Cellular Origin and Morphogenesis of the Beak

Despite the diversity and highly plastic nature of beak morphology, all avian beaks share a conservative anatomical structure covered with a horny sheath (referred to as the rhamphotheca, which includes rhinotheca and gnathotheca), consisting of the premaxilla, prenasal cartilage, and mandible (32, 33) (**Figure 2*a***). Such keratinized bony beaks also evolved in other reptile lineages, such as turtles, and nonavian dinosaurs, such as oviraptorids, which sported parrot-like beaks (34). However, birds exhibit a distinctive upper jaw characterized by elongated, narrowed, and fused premaxilla, departing from the paired premaxillae observed in the upper jaws of reptiles and ancestral archosaurs (35, 36) (**Figure 2*b***). Similar to the jaws of other vertebrates, the avian beak is a specialized structure of the craniofacial skeleton. Craniofacial development commences with the initiation of neural crest cells in the neural folds of the forebrain, midbrain, and hindbrain (37). At approximately Hamburger–Hamilton (HH) stage 9.5 of chicken embryonic development (38), neural crest cells undergo an epithelial-to-mesenchymal transition and then delaminate from the neural folds (39–41). Around HH14~HH15, neural crest mesenchyme migrate to populate four embryonic beak structures, including the frontal nasal prominence (FNP), lateral nasal prominences, maxillary prominences, and mandibular prominence (42). From HH20 to HH29, the FNP, lateral nasal prominences, and maxillary prominences grow, interact, and eventually fuse to form the upper beak, whereas the mandibular prominence develops into the lower beak (38, 43–45) (**Figure 2*c***). By HH31, the basic beak morphology is established, and in subsequent developmental stages, the beak grows proportionally with the body size of the specific bird species (38, 46).

The establishment of a species-specific beak morphology is generally considered to be determined by the skeletal elements, including prenasal cartilage and premaxillary bone, within the FNP (44), which predominantly contributes to variation in upper beak morphology (47, 48). These skeletal elements construct the foundational bony scaffold of the species-specific FNP morphology (49). Interspecific differences in FNP morphology, influenced by variations in skeletal elements, ultimately give rise to the diversity in beak morphology among birds. Comparisons of embryos of chickens (*Gallus gallus*), ducks (*Anas platyrhynchos*), and quail (*Coturnix coturnix*) with distinct beak forms demonstrate that their FNPs exhibit maximum morphological similarity at HH20 (50). However, by HH25, they begin to diverge morphologically and follow different growth trajectories. This divergence occurs because, during beak morphogenesis, neural crest mesenchyme populated within the FNP differentiate into the prenasal cartilage at HH26 and subsequently into the premaxillary bone from stages HH27 to HH30 (51) (**Figure 2*c***). Therefore, the FNP containing prenasal cartilage and premaxillary bone is of particular importance in the morphological variance of bird beaks.

2. LINEAGE-SPECIFIC GENETIC INNOVATIONS UNDERPIN BEAK DIVERSITY

Knowledge of cellular origins (i.e., neural crest cells) and determinants (i.e., FNP) in beak morphogenesis laid the groundwork for uncovering the genetic mechanisms responsible for avian beak morphogenesis and evolution. Several previous cell transplantation experiments performed by Richard A. Schneider's lab in 2003 provided compelling evidence for the role of neural crest cells in determining beak morphology and revealed that neural crest cells are a principal source of molecular information generating interspecific variation in beak morphology (reviewed in 52, 53). These experiments on transplantation of neural crest cells between duck and quail embryos showed that quail embryos transplanted with duck neural crest cells developed flat beaks resembling ducks, whereas duck embryos receiving quail neural crest cells produced donor-like beaks

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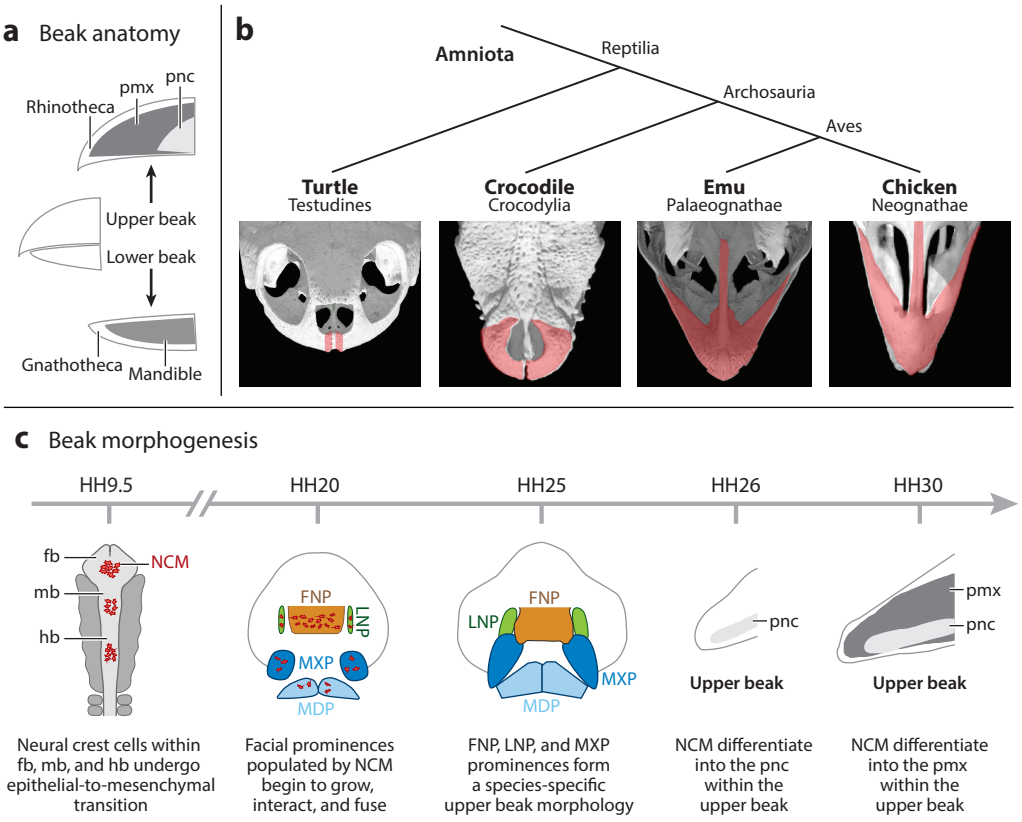


Figure 2

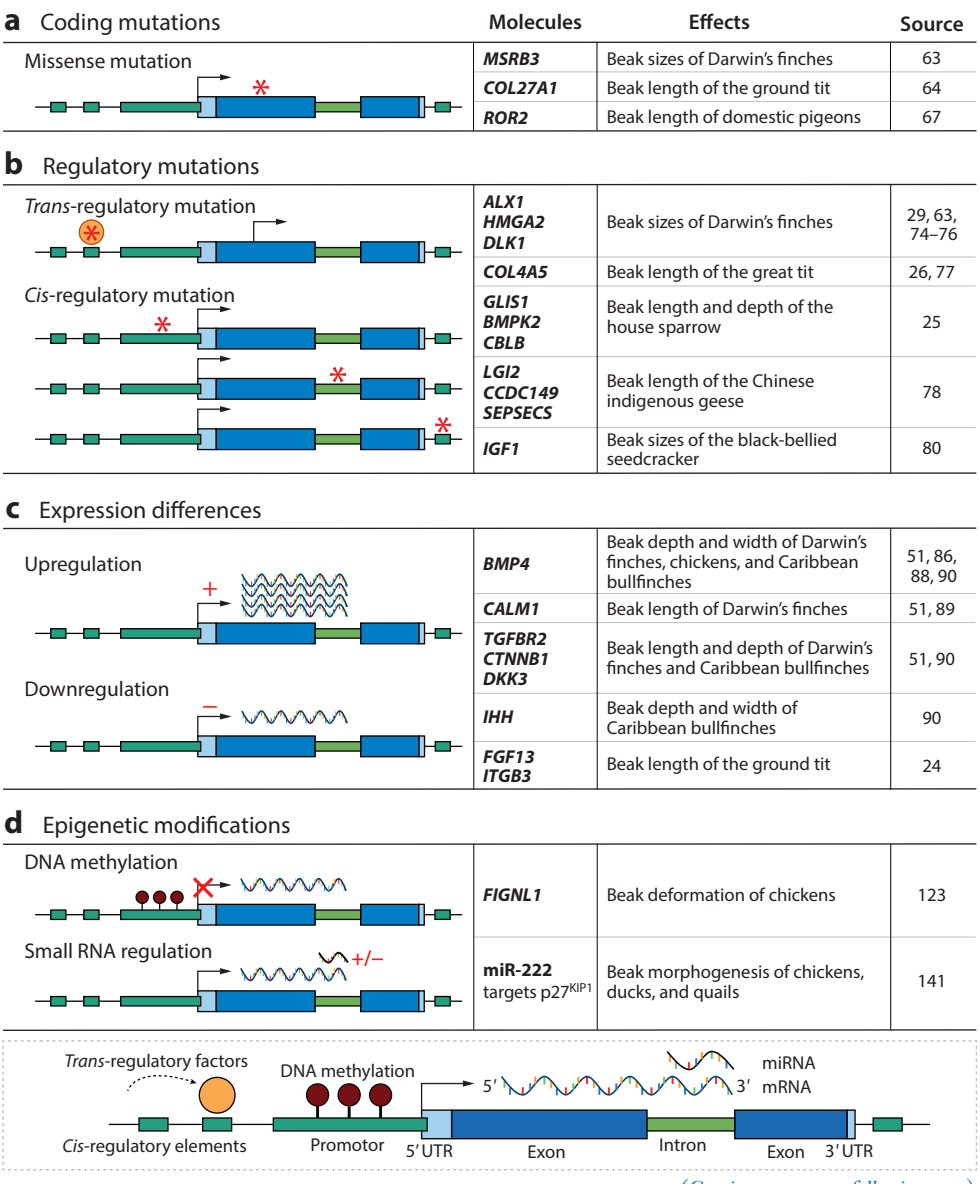
Beak anatomy and morphogenesis. (*a*) Schematic presentation of beak anatomy. The avian beak is composed of two main parts: the upper and lower beak. The upper beak's morphology is determined primarily by two bony components, the premaxilla (pmx) and prenasal cartilage (pnc), whereas the lower beak's morphology is determined by the mandible. These bony structures are covered by a horny sheath, including rhinotheca and gnathotheca. (*b*) The configuration and evolutionary origin of the premaxillae (*red shading*). The avian upper jaw is formed by elongated, narrowed, and fused pmx, which evolved from the paired pmx in reptiles and ancestral archosaurs. The skull images of the painted turtle (*Chrysemys picta*), Morelet's crocodile (*Crocodylus moreletii*), emu (*Dromaius novaehollandiae*), and domestic chicken (*Gallus gallus*) were adapted with permission from the online Digital Morphology Library at the University of Texas (**https://digimorph.org**). The amniote phylogeny and schematics of the pmx were modified with permission from Reference 36; copyright 2015 Society for the Study of Evolution. (*c*) Schematic presentation of embryos and beaks at several key stages during beak morphogenesis. The avian beak originates from neural crest cells located in neural folds of the forebrain (fb), midbrain (mb), and hindbrain (hb) that initiate the transition to mesenchyme at around Hamburger–Hamilton (HH) stage 9.5 (adapted with permission from Reference 41; copyright 2008 Company of Biologists). Subsequently, neural crest mesenchyme (NCM) populates the frontal nasal prominence (FNP), lateral nasal prominences (LNP), maxillary prominences (MXP), and mandibular prominence (MDP). At HH20, the FNP, LNP, and MXP prominences begin to grow, interact, and fuse. By HH25, these beak prominences form the species-specific upper beak morphology. Illustrations of bird beaks at HH20 and HH25 in panel *c* were created based on experimental images from Reference 50 and the style and coloring from Reference 88 with permission; copyright 2004 AAAS. This interspecific difference is derived from the pnc and pmx. Both the pnc (at HH26) and pmx (during HH27∼HH30) are formed from the NCM within the upper beak [created based on experimental images from Reference 51 (CC BY 4.0)]. All schematic drawings were created without scales.

(54–56). These studies set the direction for subsequent research on the molecular mechanisms underlying beak morphogenesis. In addition, given that the upper beak predominantly drives variation in beak morphology, with the FNP being a key contributor to upper beak variation, most current genetic mechanism research has focused on the FNP and the resulting upper beak. The

identified genetic innovations occur primarily at three levels of variation, coding mutations, regulatory mutations, and expression differences, controlling the evolution and development of beak sizes (length, height, and width) across different avian lineages (**Figure 3**).

2.1. Coding Mutations

Given that most genes regulating morphology and development exhibit pleiotropy (the expression of multiple traits by a single gene) (57), their coding sequences may be under considerable evolutionary constraints (elimination of variation due to purifying selection, also referred to as



Molecules	Effects	Source
MSRB3	Beak sizes of Darwin's finches	63
COL27A1	Beak length of the ground tit	64
ROR2	Beak length of domestic pigeons	67
ALX1 ***HMGA2*** ***DLK1***	Beak sizes of Darwin's finches	29, 63, 74–76
COL4A5	Beak length of the great tit	26, 77
GLIS1 ***BMPK2*** ***CBLB***	Beak length and depth of the house sparrow	25
LGI2 ***CCDC149*** ***SEPSECS***	Beak length of the Chinese indigenous geese	78
IGF1	Beak sizes of the black-bellied seedcracker	80
BMP4	Beak depth and width of Darwin's finches, chickens, and Caribbean bullfinches	51, 86, 88, 90
CALM1	Beak length of Darwin's finches	51, 89
TGFBR2 ***CTNNB1*** ***DKK3***	Beak length and depth of Darwin's finches and Caribbean bullfinches	51, 90
IHH	Beak depth and width of Caribbean bullfinches	90
FGF13 ***ITGB3***	Beak length of the ground tit	24
FIGNL1	Beak deformation of chickens	123
miR-222 targets $p27^{KIP1}$	Beak morphogenesis of chickens, ducks, and quails	141

(*Caption appears on following page*)

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Figure 3 (*Figure appears on preceding page*)

Identified molecular innovations underlying the evolution and development of avian beak morphology. (*a*) Mutations occur in coding sequences of genes associated with beak morphology. A missense mutation denotes a DNA change that results in different amino acids being encoded at a particular position in the resulting protein. Several genes have been shown to contain missense mutations (*red asterisk*) responsible for the evolution of beak morphology in Darwin's finches (e.g., *Geospiza* spp.), ground tits (*Pseudopodoces humilis*), and domestic pigeons (*Columba livia*). (*b*) Mutations occur in regulatory factors or sequences of genes associated with beak morphology. In contrast to coding mutations, more genes have been found to be linked to beak morphology through mutations (*red asterisk*) in regulatory factors and elements in species such as Darwin's finches, great tits (*Parus major*), house sparrows (*Passer domesticus*), Chinese indigenous geese (*Anser cygnoides*), and black-bellied seedcrackers (*Pyrenestes ostrinus*). These mutations may regulate gene expression levels, thus governing beak morphology. (*c*) Expression differences in genes responsible for beak development. Both upregulation and downregulation of several genes affect beak morphogenesis in Darwin's finches, Caribbean bullfinches (*Loxigilla* spp.), ground tits, and chickens (*Gallus gallus*). Plus and minus symbols indicate inhibition and promotion, respectively. (*d*) Epigenetic modifications and their regulators related to beak morphology. DNA methylation occurring in the promotor can lead to the decreased expression of the target gene (e.g., *FIGNL1*), thus controlling chicken beak morphology. Additionally, small RNAs such as microRNAs (miRNAs, e.g., miR-222) can posttranscriptionally regulate gene expression by binding to target messenger RNAs (mRNAs, e.g., $p27^{KIP1}$), thereby contributing to species-specific beak morphogenesis in chickens, ducks (*Anas platyrhynchos*), and quails (*Coturnix coturnix*). A schematic (*bottom*) illustrates gene structure and regulators, including exons (*blue*), 5′ and 3′ untranslated regions (UTRs) (*light blue*), introns (*light green*), *cis*-regulatory elements and a promotor (*green*), *trans*-regulatory factors (*orange*), DNA methylation (*dark red*), and single-stranded molecules (mRNA and miRNA).

conservation) (58). Once mutations arise in the highly conserved coding sequences of these genes, the structure and function of the encoded proteins would likely change (59). Such missense mutations are usually beneficial and favored by natural selection and thus can be quickly fixed and stably inherited, likely influencing morphological development and the organism's fitness (60). Accordingly, avian beaks, characterized by significant diversity in morphology, are likely influenced by coding mutations. However, only a few candidate genes have been identified as potentially linked to evolutionary variations in beak morphology (**Figure 3*a***).

Following the publication of genome assemblies for two ground finch species (61, 62), Darwin's finches have become a focus point for efforts to uncover the genetic basis of avian beak evolution, because Darwin's finches have long been a textbook example illustrating adaptive radiation associated with diverse beak morphology due to natural selection. Genomic comparisons across finch species exhibiting large, medium, and small beaks identified a considerable number of single-nucleotide polymorphisms (SNPs) with strong genetic differentiation within an ~525-kb region, with only one missense mutation occurring in the gene *MSRB3* (*methionine sulfoxide reductase B3*) (63). To assess whether the same genes or specific coding mutations also play a crucial role in beak evolution across other bird lineages, Cheng et al. (64) conducted a population genomic comparison of 14 species in the Paridae clade. Contrary to expectations, no evidence of coding mutations at the previously identified loci associated with beak morphotypes was found. Instead, two missense mutations (*R1493Q* and *P1501L*) within a highly conserved gene, *COL27A1* (*collagen type XXVII alpha 1 chain*), implicated in bone morphogenesis were predicted to exert major effects on beak elongation in the ground tit (*Pseudopodoces humilis*). This species exhibits a highly diverged beak form, a result of ecological adaptation to high-elevation open habitats and a primarily terrestrial lifestyle for ~5.7 million years, distinguishing it from all other tit species (65).

Artificial selection works in the same way as natural selection, except that artificial selection is an evolutionary process in which humans consciously select for or against a particular feature in organisms, providing a model that aids our understanding of natural selection (1). As another bird species highlighted by Darwin, domestic pigeons (*Columba livia*), under a regime

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of intensive artificial selection over thousands of years, recapitulate the principal axes of avian craniofacial morphology variation (66). One of the most striking cases of pigeon craniofacial variation is in beak morphology, observed across numerous breeds. By comparing the genomes from diverse breeds of Darwin's pigeons, Boer et al. (67) identified a missense mutation (*C1087T*) in the gene *ROR2* (*receptor tyrosine kinase-like orphan receptor 2*) as a putative regulator of beak length. These findings indicate the complexity of genetic factors influencing beak evolution across different avian lineages. However, determining whether coding mutations within these genes impact beak morphology requires additional experimental validation.

2.2. Regulatory Mutations

Although the above literature summary suggests that coding mutations induced by strong selection can drive the adaptive diversification of beak morphology within different avian lineages, adaptive changes in morphological traits are more likely attributed to mutations in *trans*-regulatory molecules (e.g., transcription factor) (68, 69) and *cis*-regulatory elements within noncoding regions (e.g., promotor, enhancer) (59, 60, 70, 71). This preference arises because more specific regulatory mutations can potentially circumvent the pleiotropic effects associated with coding mutations. Therefore, a review of the literature suggests that regulatory changes in most identified genes can account for beak morphology evolution within lineages experiencing rapid radiation or within species (**Figure 3*b***).

In the case of the rapidly radiated Darwin's finches, comparative genomic analyses between blunt- and point-beaked species have indicated a selective sweep in a transcription factor, *ALX1* (*ALX homeobox 1*) (29), associated with craniofacial development (72, 73), leading to a large number of fixed SNPs within a 240-kb region with high homozygosity in blunt-beaked species. Among these fixed SNPs, two missense mutations (*L112P* and *I208V*), which are highly conserved among birds and mammals, have been predicted to play a significant role in interspecific beak size variation among Darwin's finches (29). Notably, a severe drought in 2004–2005 led to character displacement across Darwin's finches, with large-beaked birds outcompeting small-beaked ones (4). However, this environmental shift did not result in changes in the allele frequency of *ALX1*. To identify the genetic basis responsible for the character displacement during the severe drought, further genomic comparisons among species with distinct beak sizes found more than a thousand SNPs with strong genetic differentiation across the region around *HMGA2* (*high-mobility group AT-hook protein 2*) (63). All these mutations occurred in introns and downstream and upstream regions, indicating that *HMGA2*, with two haplotypes associated with increases in beak size in post-drought medium ground finches, plays a regulatory role. A subsequent genome-wide association study (GWAS) identified two SNPs, one inside the intron of *HMGA2* and the other upstream of *DLK1* (*delta-like 1 homolog*), that were strongly linked to differences in beak morphology across deeper branches of this radiation (74). These genes were evidenced by another GWAS across this lineage based on a larger sample size (75). Recent evidence suggests that some genomic regions harboring these genes, associated with changes in beak morphology of Darwin's finches, represent ancestral haplotype blocks with origins predating speciation events during this radiation (76). Many of the genes expressed in the developing beaks, particularly *ALX1*, are overrepresented within these genomic regions.

The roles of regulatory mutations in beak variation are also likely to be found within species, particularly in those influenced by anthropogenic interference. In the case of the widespread and extensively studied great tits, mutations inside introns of the gene *COL4A5* (*collagen type IV alpha 5 chain*) have been linked to differences in beak length between populations in England and the Netherlands (26). A subsequent Europe-wide HapMap study on great tits provided evidence

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for *COL4A5* in regions with high differentiation between populations in England and Scotland (77). Both studies suggest that the elongated beaks observed in great tits in England may have evolved due to recent selection resulting from supplementary feeding practices over nearly two centuries. Such beak variation between populations has also been observed in other birds. Population genomic analysis has identified many intergenic and intronic SNPs in three BMP-related genes—*GLIS1* (*glis family zinc finger 1*), *BMPK2* (*bone morphogenic protein kinase 2*), and *CBLB* (*casitas B-lineage lymphoma proto-oncogene b*)—likely regulating beak depth and length of the house sparrow (*Passer domesticus*) (25). A combination of GWAS and quantitative genetic analyses indicated that in an economically important poultry species, the Chinese indigenous goose (*Anser cygnoides*), three candidate genes [*CCDC149* (*coiled-coil domain containing 149*), *LGI2* (*leucine-rich repeat LGI family member 2*), and *SEPSECS* (*Sep-tRNA:Sec-tRNA synthase*)] likely influence beak length development (78). All significant mutations associated with beak length within these genes are located in noncoding regulatory elements, with the exception of a missense in *SEPSECS* that does not lead to substantial change in protein structure. Additionally, as a textbook example of disruptive selection, the black-bellied seedcracker (*Pyrenestes ostrinus*) displays dramatic beak size variation within populations due to different foraging preferences for seeds with varying hardness during dry seasons (79). This beak variation is probably attributable to noncoding variants in a growth factor gene, *IGF1* (*insulin-like growth factor 1*) (80).

All these studies underscore the importance of regulatory mutations in the evolution of beak morphology across different avian lineages. A recent macroevolutionary analysis compared the genomes of 72 avian species and identified mutations in transcription factors and noncoding regions of functionally conserved genes involved in TGF-β/BMP and Wnt signaling pathways (81). These mutations are more likely than coding mutations to play vital roles in the macroevolution of avian beaks. However, it remains uncertain whether these regulatory mutations genuinely affect beak morphology by altering the expression of these conserved genes or are merely the result of linkage disequilibrium. Therefore, additional empirical studies are necessary to provide evidence for the roles of these mutations in regulating the expression of housekeeping genes or their target genes and the resulting morphological variations.

2.3. Expression Differences

In general, morphological differences in developing beaks between two species with distinct adult beak forms typically can be observed during beak morphogenesis spanning stages from HH26 to HH30 (50, 82). This development phase closely follows the phylotypic stage that is characterized by high morphological and molecular resemblance among different species' embryos (83). Morphological development is more likely influenced by alterations in the expression of responsible genes rather than their coding mutations (60). Therefore, in the pre-genome-sequencing era, researchers relied on quantitative genetics and microarrays, which interrogate single loci or a relatively small number of genes compared to genome-wide transcriptomic analyses, to measure the effects of gene expression differences on beak morphogenesis. Before that, craniofacial development had already been associated with the expression activities of numerous growth factors (53). Among these factors, two important candidate genes, *FGF8* (*fibroblast growth factor 8*) and *SHH* (*sonic hedgehog*), exhibit high levels of expression in the primordial embryonic beak of chickens, which appears to drive cartilage outgrowth before beak morphogenesis (84, 85). In a detailed study of beak development in Darwin's finches, however, *FGF8* and *SHH* did not seem to modulate beak morphology (86). This observation may be because these two genes primarily regulate the transformation from an ancestral beak with paired, abbreviated, and rounded premaxillae to an avian beak with elongated, narrowed, and fused premaxilla (36, 87), rather than regulating

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variations in beak size and shape among different lineages or species. Instead, the expression of *BMP4* (*bone morphogenetic protein 4*) was found at higher levels in all ground finches at HH26, which have deeper and broader beaks compared to other finch species with shallow and narrow beaks (86). Experimental mis-expression of *BMP4* in the FNP of embryonic chickens resulted in a larger upper beak (86, 88). These two studies by Abzhanov et al. (86) and Wu et al. (88), which simultaneously examined the regulatory roles of *BMP4* in beak morphogenesis, represent a significant breakthrough in exploring the molecular mechanisms underlying beak morphology (**Figure 3*c***).

However, these studies demonstrate the pivotal role of *BMP4* in regulating beak depth and width rather than length, at least in chickens and Darwin's finches. A subsequent microarray analysis in Darwin's finch embryos revealed an association between *CALM1* (*calmodulin 1*), a gene involved in the CaM-dependent pathway, and differences in beak length (89). Overexpression of *CALM1* in the FNP results in chicks with an elongated upper beak. Notably, during the embryonic development of Darwin's finches with varied beak morphology, *BMP4* and *CALM1* jointly regulate the morphogenesis of the early-developed prenasal cartilage at HH26, whereas other three genes [*TGFBR2* (*transforming growth factor beta receptor 2*), *CTNNB1* (*catenin beta 1*), and *DKK3* (*dickkopf 3*)] are differentially expressed in embryonic upper beaks and jointly govern the depth and length of the late-developed premaxillary bone at HH27∼HH30 stages (51). These two molecular modules, operating at different phases of beak morphogenesis, combine to determine multidimensional variations in beaks of Darwin's finches. This mechanism is also observed in one of three *Loxigilla* species closely related to Darwin's finches, whereas in the other two, *IHH* (*Indian hedgehog*) can regulate the expansion of the premaxillary bone only by binding to *BMP4* (90). In a more distantly related songbird lineage (Paridae), a comparative transcriptomic analysis between long-beaked ground tits and short-beaked great tits provides evidence that shifts in the expression of *FGF13* (*fibroblast growth factor 13*) and *ITGB3* (*integrin beta 3*) respectively modulate osteoblast and osteoclast concentration in the developing FNP at HH27/28 to regulate upper beak length (24).

Current gene expression studies suggest that the same molecular module may shape different morphs, and conversely, the congruent morph may also be generated by different molecular modules. This complexity underscores the intricate interplay of genetic factors shaping avian beak development. Furthermore, it remains crucial to investigate if the expression of these genes associated with beak morphology might be the result of regulatory mutations in their noncoding regions or secondary effects resulting from coding mutations in other genes.

3. EPIGENETIC INNOVATIONS MAY FUEL BEAK PLASTICITY AND ADAPTIVE CHANGE

When birds confront rapid environmental changes, such as drought-induced food shortages or alterations in habitat due to urbanization, their beak morphology may undergo plastic changes under environmental pressures. In such scenarios, there might not be sufficient time for novel genetic variations to emerge. Instead, alterations in the expression of genes harboring standing genetic variation may be responsible for these plastic changes, as found in sticklebacks (91), although genetic variation, whether in coding or noncoding regions, can maintain the major morphological space of the beak within a lineage. However, the regulation and maintenance of the expression of these responsible genes under such circumstances are not yet fully understood. Currently, it is widely accepted that transcription factors binding to *cis*-regulatory sequences (e.g., promotor or enhancer) coordinate the regulation of gene expression (92). In this process, nongenetic instead of genetic variations occurring in *cis*-regulatory regions near a responsible gene—although

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both can cause alteration of gene expression—can respond to environmental stimuli more quickly and efficiently. Epigenetic modifications (e.g., DNA methylation, small RNA regulation, histone modification, and chromatin remodeling) are a category of such heritable changes (93, 94) that can regulate gene expression without altering the DNA sequences (95). A growing body of evidence in animals and plants suggests that epigenetic modifications play crucial roles in regulating phenotypic plasticity and facilitating rapid evolutionary adaptation to environmental changes (96–105).

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3.1. DNA Methylation

Among epigenetic modifications, DNA methylation is the most extensively understood epigenetic regulatory mechanism of gene expression (98). Hypermethylation in the promotor can lead to decreased expression of target genes by modifying chromatin states (106) or blocking transcription factor binding (107). DNA methylation involves the addition of a methyl group (–CH3) at the C5 position of cytosine (5-methylcytosine) (96). It can be epigenetically heritable through mitotic cell divisions (108, 109) or transgenerationally heritable at certain loci (110, 111). Therefore, DNA methylation is conserved in somatic tissues and can persist over evolutionary time, facilitating its potential roles in the evolution and development of these complicated traits (110, 112–114). The role of DNA methylation in bone development has also been documented in humans and rats (115, 116). However, these studies have focused on mammals; the function of DNA methylation in birds remains poorly investigated, limiting our understanding of its effects on the evolution of diverse avian phenotypes.

Like in mammals, DNA methylation predominantly targets CpG dinucleotides in birds (117, 118). Two genome-wide studies in great tits support the idea that DNA methylation is not randomly distributed across the genome (119, 120). Instead, it disproportionately occurs around transcription start sites in the promotor regions of genes largely responsible for brain development. Similarly, across Darwin's finches, DNA methylation is also located primarily in numerous genes likely related to phenotypic differentiation, e.g., beak size and plumage coloration, resulting in population differentiation driven by environmental changes, such as urbanization (121, 122). Recently, a study integrating whole-genome bisulfite sequencing and RNA sequencing data has revealed the involvement of the methylated *FIGNL1* (*fidgetin-like protein 1*) gene in the mandibular malformation of chicken beaks (123) (**Figure 3*d***).

3.2. Small RNA Regulation

As epigenetic modulators, microRNAs (miRNAs), short noncoding RNAs typically ranging from 18 to 25 nt, exert regulatory control over gene expression posttranscriptionally by binding to the untranslated regions of target messenger RNAs (mRNAs) (124–126). Because each miRNA can potentially target hundreds of mRNAs (127), miRNAs could significantly impact the profiles of transcriptomes and proteomes in eukaryotes (128, 129), thereby influencing a broad spectrum of biological processes and phenotypic traits. Variations in miRNAs or their target sequences contribute to rapid phenotype diversification, as observed in cichlids (130, 131). Moreover, shifts in miRNA expression may play significant roles in human and animal development, as extensively reviewed in References 132 and 133, particularly in processes crucial for craniofacial development in mice, such as cellular proliferation, differentiation, and epithelial–mesenchymal transitions (134–140). However, these effects have scarcely been reported in birds. A comparative analysis of miRNA expression levels between neural crest cells in the FNP from chickens, ducks, and quails before and after species-specific facial distinctions occurred has identified five differentially expressed miRNAs (e.g., miR-222) with high conservation within all avian species (141).

These miRNAs target specific mRNAs, e.g., the cell cycle regulator $p27^{KIP1}$ encoded by the gene *CDKN1B* (*cyclin-dependent kinase inhibitor 1B*), involved in the Wnt signaling pathway, thereby contributing to species-specific beak morphogenesis in ducks, chickens, and quails (**Figure 3*d***).

These studies underscore the significant role of epigenetic modifications in influencing the adaptive changes in phenotypes of organisms, particularly during the initial responses to environmental shifts. Thus, the effects of epigenetic variations on adaptive alterations and plasticity of beaks in birds must be investigated. This exploration is particularly crucial when comparing populations within a species or different species within a lineage experiencing rapid radiation, such as Hawaiian honeycreepers. In this exploration, the use of epigenomic methods, such as whole-genome bisulfite sequencing, transposase-accessible chromatin with sequencing (142), and chromatin immunoprecipitation sequencing (143), is essential. To date, despite several attempts to investigate the epigenetic regulatory mechanisms of beak morphology in domestic birds, no instances have been observed in wild bird populations.

Of note, the red crossbill (*Loxia curvirostra*) has evolved a specialized beak morphology in which the lower beak crosses to either the right or the left over the upper beak, aiding in breaking pine cone scales and extracting seeds. Intriguingly, crossbill nestlings and fledglings initially possess a wedge-shaped beak like other finches (144). As they independently feed on seeds from tough cones, their beaks undergo rapid growth, ultimately adopting a fully crossed configuration. These observations indicate that crossed beaks in red crossbills are not born with them and cannot be observed during the morphogenesis process. After reaching adulthood, red crossbills exhibit both left- and right-crossing beak morphs with equal frequency. An empirical study that tested a hypothesis of frequency-dependent selection supported this 1:1 ratio (145). We postulate that the crossing type of crossbills' beaks may hinge on cone scale orientation during early-life feeding; if the cone scale orients to the left, their lower beak crosses to the right, and vice versa. This post-adult environment-dependent interindividual difference in crossbill beak morphology is hypothesized to be driven predominantly by epigenetic variations, given the environmental susceptibility and spatiotemporal regulation characteristic of epigenetic modifications.

4. CONCLUDING REMARKS

Studies based on morphology, developmental biology, quantitative genetics, transcriptomics, genomics, and epigenomics show that the molecular mechanisms underlying avian beak evolution and development are largely regulatory and evidently complicated. This complexity indicates that the beak, with its diverse and highly variable nature, aligns with the characteristics typical of a polygenic trait, as commonly observed in many quantitative phenotypes (146). According to the polygenic model, natural selection acts on multiple loci associated with various aspects of beak morphology simultaneously (147). This results in the joint role of a few major-effect loci alongside extensive modulation by small-effect loci in controlling adaptive changes in beak morphology (148). Although the examination of only a few bird lineages limits the generalizability of these findings, the available evidence suggests that major-effect loci are often lineage specific. In other words, distinct beak morphologies within a lineage may arise from unique molecular mechanisms. The between-lineage variations in molecular mechanisms may be attributed to different ecological selective pressures or their varying strength across lineages. Additionally, differences in analytical approaches or data sizes in different studies could also lead to these variations. Despite the lineage-specific nature of these major-effect loci, all identified genes or regulators tend to align with similar functions related to mesenchymal cell proliferation, cartilage development, and bone morphogenesis, most of which are involved in the TGF-β/BMP, Wnt, FGF/MAPK, calcium, Notch, Hippo, and Hedgehog signaling pathways as well as osteoclast

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differentiation. This suggests that the interaction and synergism among different genes or regulatory factors likely play a crucial role in regulating beak evolution and development.

5. FUTURE PROSPECTS

Avian beak morphology has long captivated evolutionary biologists, spurring extensive research into the molecular mechanisms governing its evolution and development over the past 20 years. The ongoing advancements in omics technologies and embryo manipulation methods hold promise for delving even deeper into this intriguing topic. However, summarizing the findings of previous studies, certain limitations must be acknowledged. We thus conclude with several recommendations for future research in this classic and fascinating field.

5.1. Study a Wider Array of Avian Lineages

Among the more than 11,000 extant bird species, nearly every major clade possesses its own unique beak morphology. Even within each clade, beak morphology varies across multiple dimensions, with the passerines showcasing particularly noteworthy diversity and plasticity (**Figure 1**). Despite the diverse range of beak forms among birds, only a very small fraction has been studied so far, focusing mainly on poultry and a dozen wild bird species (**Figure 3**). This greatly limits our understanding of the molecular mechanisms underlying beak evolution and development. Hence, future effort must focus on conducting comprehensive studies for bird lineages with specialized or diverse beak forms that remain unexplored but that would be easy to study, such as flat-beaked spoonbills, hook-beaked parrots, sword-beaked hummingbirds, and curve-beaked scimitar babblers. Such endeavors promise to yield more valuable insights into the molecular basis of beak evolution and development. By integrating molecular mechanisms across different lineages with varied beak forms, researchers can delve into their evolutionary origins and test the universality of lineage-specific regulatory mechanisms among birds. Furthermore, by conducting a comparative genomic analysis among birds, reptiles, and mammals, researchers can also identify the underlying molecular mechanisms responsible for the evolutionary transition of the avian fused premaxilla from the reptile-like paired premaxillae.

5.2. Increased Attention to Beak Shape

Variations in beak morphology may occur in terms of size, shape, or both. However, almost all current studies have focused on elucidating the molecular regulation mechanisms of variations in beak sizes (i.e., length, width, height) rather than beak shapes. This gap may arise from the complexity inherent in comparative studies of beak shapes, which necessitate consideration of variations in multiple dimensions, such as curvature, symmetry, directionality, and architecture (149). Compared to beak sizes, the molecular regulatory system governing beak shapes may be more intricate, posing a significant challenge in investigating their regulatory mechanisms. With increased focus on additional lineages and innovations in multi-omics joint analysis methods, this challenge can be overcome. Given that the early diversification process of beak morphology occurred primarily in beak shapes, unraveling the underlying molecular mechanisms of beak shapes is crucial for elucidating the evolutionary mechanisms of beak morphology. Therefore, more attention should be paid to lineages with highly specialized beak shapes, such as the flamingos, hummingbirds, and crossbills mentioned above.

5.3. Expanding Efforts on the Lower Beak

The notion that the upper beak governs the morphology of the entire beak has long guided ecological, developmental, and genetic studies, tracing back to Darwin's era. Consequently, studies

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have often employed upper beak morphology as a proxy for overall beak morphology. However, the lower beak, equally vital for bird feeding, has frequently been overlooked. The lower beak possesses anatomical structures and morphogenetic processes independent from the upper beak (**Figure 2**), suggesting potentially different molecular regulations. Several studies have extensively explored the developmental and molecular mechanisms regulating the species-specific morphology of the lower beak (41, 150–152), revealing distinct mechanisms compared to those governing the upper beak. Although a commentary article has proposed a potential coordinating role of the lower beak in upper beak morphology (153), understanding how this coordination operates during developmental and evolutionary processes requires further investigation. Expanding research efforts to study the lower beak will elucidate the molecular intricacies shaping avian beak development and evolution.

5.4. Identifying Additional Molecular Mechanisms

In addition to aforementioned innovations, additional molecular mechanisms, such as genomic structure variation, alternative splicing, histone modification, and long noncoding RNA, may also play pivotal roles in the evolution and development of such a complex polygenic trait. Further studies using advanced sequencing technologies and analysis tools involved in both genomics and epigenomics should be performed across a broader spectrum of bird lineages. So far, genomes for more than 1,100 bird species have been released in the National Center for Biotechnology Information, with gene annotation information available for 386 species (**https://www.ncbi.nlm.nih.gov/**, accessed Aug. 8, 2024). These annotated assemblies can serve as valuable resources for identifying structure variations across numerous species with distinct beak sizes and shapes (154) and can also provide references for other types of sequencing data, e.g., whole-genome resequencing, RNA sequencing, and chromatin immunoprecipitation sequencing, to uncover corresponding variations potentially associated with beak morphology. In particular, the use of two cutting-edge techniques, single-cell RNA sequencing and spatial transcriptomics, holds promise in elucidating species-specific molecular information in neural crest cells during early embryo development. In the future, incorporating these potential additional mechanisms, alongside the identified regulatory molecules and pathways, will be instrumental in constructing a comprehensive blueprint for understanding the molecular mechanisms that control avian beak development and morphological diversification.

5.5. Conducting Functional Validation Experiments

Early quantitative genetic studies on beak morphogenesis typically involved some gain-of-function or loss-of-function experiments. However, candidate genes, small molecules, or molecular variations potentially relevant to avian beak evolution, identified through transcriptomic, genomic, and epigenomic approaches, still lack direct evidence for functional significance. Given the potential for false positives in certain candidate molecules identified through omics studies, functional validation for their variations must be conducted. So far, a series of state-of-the-art experimental technologies, including RNA interference, gene editing, and epigenetic editing, have been widely used for functional validation of variations in the expression and sequences of molecules. In particular, the CRISPR/Cas9 (clustered regularly interspaced short palindromic repeats–associated protein 9) editing system is maturing in both somatic and germ cell lineages of chickens (155). Remarkably, the newly launched AlphaFold 3 (156) will provide better assurance for functional experiments. Prior to conducting experiments, AlphaFold 3 can be used to predict structural and functional alterations induced by mutations, as well as predict other proteins, nucleic acids, and small molecules that potentially interact with candidate molecules (156). In this

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case, use of these tools would not only validate the functions of identified genes or variations but also reproduce any required potential beak morphs in model species embryos.

In summary, despite growing researcher engagement and abundant genomic and epigenomic data, exploring the molecular mechanisms governing the evolution and morphogenesis of bird beaks remains as challenging as ever. One obstacle stems from the ethical considerations and wildlife conservation issues surrounding the collection of wild bird embryos for evolutionary developmental biology research. Addressing this challenge requires strategic collaborations between research institutions and wildlife conservation and animal ethics management bodies. The second roadblock is that although genomes of more than 1,100 species are available, most are of low quality, and only a small portion of chromosome-level assemblies can be used to detect genomic structural variation associated with beak morphology. Applying third-generation/long-read sequencing technologies will produce more high-quality chromosome-level genomes. Additionally, effectively processing and integrating the increasing volume of multi-omics data poses another significant challenge. Fostering academic partnerships among researchers from various fields, such as ecology, evolutionary biology, developmental biology, cell biology, genomics, epigenomics, bioinformatics, and statistics, is essential for finding innovative solutions.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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